

JK BEGUMISA RESOURCES

IDENTITY: DIFFERENTIATION I & II – REVISION QUESTIONS

DATE: 21/10/2025.

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NOTES:

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DIFFERENTIATING FROM FIRST PRINCIPLES:

i). Differentiate the following from first principles.

1. $\cos^2\left(\frac{2x}{3}\right)$

2. $y = (2x^2 + 5x)$

3. $f(x) = \ln 2x$

4. $y = \sin x$

5. $y = \frac{2}{3x^2-3}$

6. $f(x) = \operatorname{cosec} x$

7. $h(x) = \cot x$

8. $g(x) = \tan^2 2x$

9. $y = \sec\left(\frac{x}{2}\right)$

10. $y = 2x^{\frac{1}{2}} - 3$

11. $f(x) = \cos^2\left(\frac{x}{2}\right)$

12. $y = \sin\left(\frac{x}{4}\right)$

13. $y = \sec^2 x$

14. $h(x) = \tan\left(\frac{x}{2}\right)$

15. $y = \frac{1}{2\sqrt{x+3}}$

16. $f(x) = (3x^2 + 4x) \sin(5x + 9)$

17. $y = \frac{10}{\sqrt{4x^2+3}}$

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ii). In each of the following, show that;

18. $\frac{\delta y}{\delta x} = \frac{1}{\sqrt{(x+\delta x)}+\sqrt{x}}$ given that; $y = \sqrt{x}$. Hence, deduce; $\frac{dy}{dx}$.

19. $\frac{\delta y}{\delta x} = \frac{1}{\left((x+\delta x)^{\frac{1}{4}+x^{\frac{1}{4}}}\right)\left((x+\delta x)^{\frac{1}{2}+x^{\frac{1}{2}}}\right)}$ given that $y = x^{\frac{1}{4}}$. Hence, deduce; $\frac{dy}{dx}$.

20. $\frac{\delta y}{\delta x} = -\frac{\sin\left(x+\frac{\delta x}{2}\right)}{\sqrt{\cos(x+\delta x)}+\sqrt{\cos x}}$ given that $y = \sqrt{\cos x}$. Hence, deduce; $\frac{dy}{dx}$.

21. $\delta y = \frac{-\sin(2\delta x)}{\sin 2(x+\delta x)\sin 2x(\sqrt{\cot 2(x+\delta x)}+\sqrt{\cot 2x})}$ given that $y = \sqrt{\cot 2x}$. Hence, deduce; $\frac{dy}{dx}$.

22. $\delta y = 2 \cos(2x + \delta x) \sin \delta x$ given that $y = \sin 2x$. Hence, deduce; $\frac{dy}{dx}$.

USE OF RULES OF DIFFERENTIATION:

iii). Differentiate the following w. r. t. x, giving your answers in the simplest form.

23. $\ln \sqrt{\frac{1+x}{1-x}}$

35. $\log \sqrt{\frac{1-\sin x}{1+\sin x}}$

24. x^x

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36. $\cos\left(\log_2(4x^2-2x)\right)$

25. $\frac{x^3-1}{x^2}$

37. $\sec(\tan \sqrt{x})$

26. $\frac{6x+7}{x^2-x+3}$

38. $\frac{e^x+\log x}{\sin 3x}$

27. $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$

39. $\frac{x^2+\sin x}{\cot x}$

28. $4^{3x} \sin\left(\frac{x}{2}\right)$

40. $\frac{1}{\log(\cos x)}$

29. $8x^{\cos\left(\frac{x}{3}\right)}$

41. $\cos\left(\frac{x}{x+1}\right)$

30. $(2x)^{\sin\left(\frac{3x}{2}\right)}$

42. $x \log(3x)$

31. $x \tan^{-1}(2x)$

43. $\frac{\sin^2 x}{1+\cos^2 x}$

32. $x^{\frac{1}{2}} \ln(3x)$

44. $\frac{x}{\sin 3x}$

33. $\frac{1-10x}{(2x-1)^5}$

45. $\sqrt{\tan \sqrt{x}}$

34. $x = 3 \tan 2y$

46. $\sin(\sqrt{\sin x + \cos x})$

47. $\log\left(\frac{1+x}{1-x}\right)$

48. $\sin(e^x \log 5x)$

49. $xy^3 = 2\ln y$

50. $\ln x - 2\ln y + 5xy = 3$

51. $\ln(xy) = 4 - y^2$

52. $2ye^{4x} + e^{x^2}y^3 = 10$

53. $x\ln y = 2x + 5$

54. $x^3 + 2xy + y^3 = 10$

55. $\log_7^{(\log x)}$

56. $e^{4x} \sin 3x$

57. $x \sin^{-1}(x-1)$

58. $\tan^{-1}(\ln x)$

59. $2^{(3+\sin x)}$

60. $4^{-x^2} \cos x$

61. $10^{\ln 2x}$

62. $s(t) = \cos(2^{3t})$

63. $\left(\frac{3}{2}\right)^x \ln\left(\frac{3}{2}\right)$

64. $\cos^{-1}(\sqrt{x})$

65. $\ln \sqrt{\frac{3+2x}{3-2x}}$

66. $\sin(x^x)$

67. $4^x \sin x$

68. $\ln \left\{ e^x \left(\frac{x-2}{x+2} \right) \right\}^{\frac{3}{4}}$

69. $e^{x+4y} \cos(2^x)$

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PROOFS IN DIFFERENTIATION:

70. Show that; $\frac{d}{dx} \left(\log_e^{(x+\sqrt{x^2-1})} \right) = \frac{1}{\sqrt{x^2-1}}$

71. Show that; $\frac{d}{dx} \left(\frac{\log x}{x} \right) = \frac{1}{\ln 10} \left(\frac{1-\ln x}{x^2} \right)$

72. If $y = e^{mx} + e^{-mx}$, show that; $\frac{d^2}{dx^2}(y) = m^2 y$

73. Given that; $x = \sec^2 3y$,

i) Show that; $\frac{dy}{dx} = \frac{1}{6x(x-1)^{\frac{1}{2}}}$

ii) Hence, find; $\frac{d^2 y}{dx^2}$

74. Show that; $\frac{d^2}{dx^2} (\log x + a^x) = -\frac{1}{x^2 \ln 10} + a^x (\ln a)^2$

75. Given that; $y = \log \left(\frac{\sqrt{x+a} + \sqrt{x-a}}{\sqrt{x+a} - \sqrt{x-a}} \right)$, show that; $\frac{dy}{dx} = \frac{1}{\ln 10 (\sqrt{x^2-a^2})}$

76. Show that; $\frac{d}{dx} \left(\sqrt{\frac{1+\sin x}{1-\sin x}} \right) = \sec x (\sec x + \tan x)$
77. Given that; $y = \tan xy$, show that; $\frac{dy}{dx} = \frac{y}{\cos^2 xy - x}$
78. Show that; $\frac{d}{dx} (\tan^{-1}(x^x)) = \frac{(1+\ln x)}{(x^{-x}+x^x)}$
79. Show that; $\frac{d}{dx} \left(\frac{x+1}{\sqrt{(x^2-1)}} \right) = \frac{-1}{(x-1)\sqrt{(x^2-1)}}$
80. Given that; $\sqrt{1-x^2} + \sqrt{1-y^2} = m(x-y)$ where m is a constant, show that;

$$\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$$
81. Given that; $x^m y^n = (x+y)^{m+n}$, prove that; $\frac{dy}{dx} = \frac{y}{x}$.
82. Given that; $x = e^{\sec y}$, show that; $\frac{dy}{dx} = \frac{1}{x \ln x (\sqrt{(\ln x)^2 - 1})}$.
83. If $y = e^{-x} \cos x$, prove that; $\frac{dy}{dx} = -\sqrt{2} e^{-x} \cos \left(x - \frac{\pi}{4} \right)$
84. If $y = x^x$, prove that; $\frac{d^2 y}{dx^2} - \frac{1}{y} \left(\frac{dy}{dx} \right)^2 - \frac{y}{x} = 0$
85. If $y = (x + \sqrt{1+x^2})^n$, show that; $(1+x^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = n^2 y$
86. If $y = P e^{ax} + Q e^{bx}$, show that; $\frac{d^2 y}{dx^2} - (a+b) \frac{dy}{dx} + aby = 0$
87. If $y = \log(x + \sqrt{x^2 + a^2})$ where a is a constant, show that; $(x^2 + a^2) \frac{d^2 y}{dx^2} + x \frac{dy}{dx} = 0$
88. If $y = \sin(\ln x)$, then prove that; $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$
89. If $y = \sin^{-1}(x)$, show that; $(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} = 0$
90. A curve is given by; $xy^n = \beta$ where β, n are all constants, prove that; $\frac{d^2 y}{dx^2} = \frac{y(1+n)}{n^2 x^2}$
91. If $y = A e^{3x} + B e^{-2x}$, prove that; $\frac{d^2 y}{dx^2} - \frac{dy}{dx} - 6y = 0$
92. If $y^2 - 2xy = 2x$, prove that; $(x-y) \frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} = \left(\frac{dy}{dx} \right)^2$
93. If $e^x y = \sin x$, prove that; $\frac{d^2 y}{dx^2} + 2 \frac{dy}{dx} + 2y = 0$
94. If $y = a e^{-2x} \sin 3x$, prove that; $\frac{d^2 y}{dx^2} + 4 \frac{dy}{dx} + 13y = 0$
95. If $y = \frac{\sin^{-1}(x)}{\sqrt{1-x^2}}$, prove that; $(1-x^2) \frac{d^2 y}{dx^2} - 3x \frac{dy}{dx} - y = 0$
96. If $y = 3 \cos(\ln x) + 4 \sin(\ln x)$, then show that; $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + y = 0$
97. If $y = e^{a \sin^{-1}(x)}$ for $-1 \leq x \leq 1$ and a is a constant, then show that;

$$(1-x^2) \frac{d^2 y}{dx^2} - x \frac{dy}{dx} - a^2 y = 0.$$
98. If $x = \tan \left(\frac{1}{a} \ln y \right)$, then show that; $(1+x^2) \frac{d^2 y}{dx^2} + (2x-a) \frac{dy}{dx} = 0$

99. If $y = \operatorname{cosec}^{-1}(x)$, then show that; $x(x^2 - 1) \frac{d^2y}{dx^2} + (2x^2 - 1) \frac{dy}{dx} = 0$

100. If $y = x \ln \left(\frac{x}{a+bx} \right)$, prove that; $x^3 \frac{d^2y}{dx^2} = \left(x \frac{dy}{dx} - y \right)^2$

101. If $y^x = e^{y-x}$, prove that; $\frac{dy}{dx} = \frac{y^2}{x(y-x)}$

Hence, deduce that; $(xy - x^2) \frac{d^2y}{dx^2} + x \left(\frac{dy}{dx} \right)^2 = (y + 2x) \frac{dy}{dx}$.

102. If $x^y = e^{x-y}$, prove that; $\frac{dy}{dx} = \frac{y(x-y)}{x^2}$.

Hence, deduce that; $x^2 \frac{d^2y}{dx^2} + (2y + x) \frac{dy}{dx} = y$

103. If $\log_e^{(x^2+y^2)} = 2 \tan^{-1} \left(\frac{y}{x} \right)$, show that; $\frac{dy}{dx} = \frac{x+y}{x-y}$.

Hence, deduce that; $(x - y) \frac{d^2y}{dx^2} - \left(\frac{dy}{dx} \right)^2 = 1$

104. If $y = (\cot^{-1} x)^2$, show that; $(1 + x^2)^2 \frac{d^2y}{dx^2} + 2x(1 + x^2) \frac{dy}{dx} = 2$

105. If $y = \frac{1-\sin 2x}{1+\sin 2x}$, show that; $\frac{dy}{dx} + 2 \tan \left(\frac{\pi}{4} - x \right) \sec^2 \left(\frac{\pi}{4} - x \right) = 0$.

106. Prove that; $\frac{d}{dx} \left(\frac{x}{2} \sqrt{a^2 - x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} \right) = \sqrt{a^2 - x^2}$

107. If $e^y(x+1) = 1$, show that $\frac{d^2y}{dx^2} = \left(\frac{dy}{dx} \right)^2$

108. If $y = x + \tan x$, show that; $\cos^2 x \frac{d^2y}{dx^2} - 2(y-x) = 0$

109. If $y = \log \sqrt{\frac{1-\cos x}{1+\cos x}}$, show that; $\frac{dy}{dx} = \frac{1}{\ln 10} \operatorname{cosec} x$

110. If $y = e^x(\sin x + \cos x)$, prove that; $\frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + 2y = 0$

111. If $y = \sqrt{\frac{1-x}{1+x}}$, show that; $(1-x^2) \frac{d^2y}{dx^2} = (2x-1) \frac{dy}{dx}$

112. If $y = \sqrt{x} + \frac{1}{\sqrt{x}}$, then show that; $2x \frac{dy}{dx} + y = 2\sqrt{x}$.

Hence, deduce that; $2x \frac{d^2y}{dx^2} + 3 \frac{dy}{dx} + \sqrt{x} = y$

113. If $y = (\ln x + \sqrt{1+x^2})^2$, then show that; $(1+x^2) \frac{d^2y}{dx^2} + x \frac{dy}{dx} - 2 = 0$

114. If $y = \frac{1}{\sqrt{x}}$, show that; $2x \frac{d^2y}{dx^2} + 3 \frac{dy}{dx} = 0$

115. If $e^x = \tan 2y$, show that; $\frac{d^2y}{dx^2} = \frac{e^x - e^{3x}}{2(1+e^{2x})^2}$.

116. If $y = \frac{\ln(1+x)}{x^2}$, show that; $x^2 \frac{dy}{dx} + 2xy = \frac{1}{1+x}$.

117. If $y = \ln(\ln x)$, show that; $\ln x \frac{d^2y}{dx^2} + \frac{1}{x} \frac{dy}{dx} + \frac{1}{x^2} = 0$.

118. If $y = 2 \sin 2x \ln(\tan x)$, show that; $\frac{d^2y}{dx^2} + 4y = 8 \cot 2x$

119. If $y = \sqrt{1 + \sin 2x}$, show that; $\frac{dy}{dx} = \sqrt{1 - \sin 2x}$.

Hence show that; $\frac{d^2y}{dx^2} + y = 0$.

120. If $y = x\sqrt{1+x}$, show that; $\frac{dy}{dx} = \frac{(3x+2)}{2\sqrt{1+x}}$.

121. If $y = \tan^{-1}(1-x)^{\frac{1}{2}}$, show that; $(2-x)\frac{d^2y}{dx^2} - \frac{dy}{dx} + \frac{1}{4}(1-x)^{-\frac{3}{2}} = 0$.

122. If $y = \sec x + 2\tan x$, show that; $\cos x \frac{dy}{dx} + 3\tan x = 2y$.

Hence, show that; $\cos x \frac{d^2y}{dx^2} - (2 + \sin x) \frac{dy}{dx} + 3 \sec^2 x = 0$

123. If $x = \ln(xy)$, show that; $\frac{dy}{dx} = \left(\frac{x-1}{x^2}\right)e^x$.

Hence, show that; $x \frac{d^2y}{dx^2} + (2-x) \frac{dy}{dx} = y$

124. If $y = ae^x + b\cos x$, show that; $(1 + \tan x) \frac{d^2y}{dx^2} - 2 \frac{dy}{dx} + (1 - \tan x)y = 0$.

125. If $y = \tan^{-1}(\log_e x^2)$, show that; $x \frac{d^2y}{dx^2} + \frac{dy}{dx} + 2\sin 2y = 0$.

126. If $u = \theta^2 + (\sin^{-1} \theta)^2 - 2\theta\sqrt{1-\theta^2} \sin^{-1} \theta$,
show that; $\sqrt{1-\theta^2} \frac{du}{d\theta} = 4\theta^2 \sin^{-1} \theta$.

127. If $y = \frac{\sin x}{x^2}$, show that; $x^2 \frac{d^2y}{dx^2} + 4x \frac{dy}{dx} + (x^2 + 2)y = 0$.

128. If $y = \sin^{-1}(3x - 4x^3)$, show that; $\sqrt{1-x^2} \frac{dy}{dx} = 3$.

Hence, deduce that; $(1-x^2) \frac{d^2y}{dx^2} = x \frac{dy}{dx}$.

129. If α and β are constants, show that the derivative with respect to x of each of the functions;

i) $2 \sin^{-1} \sqrt{\frac{x-\beta}{\alpha-\beta}}$

ii) $2 \tan^{-1} \sqrt{\frac{x-\beta}{\alpha-\beta}}$

is; $((\alpha - x)(x - \beta))^{\frac{1}{2}}$

130. Show that if; $u = \tan^{-1}(\theta)$, then; $(1 + \theta^2) \frac{d^2u}{d\theta^2} + 2\theta \frac{du}{d\theta} = 0$

131. If $y = e^{\sin 2x}$, show that; $\frac{d^2y}{dx^2} = \frac{1}{2}y(1 + 4\cos 2x - \cos 4x)$.

132. If $y = \tan^{-1}\left(\frac{ax-b}{bx+a}\right)$, show that; $\frac{d^2y}{dx^2} + 2x\left(\frac{dy}{dx}\right)^2 = 0$.

133. Show that; $\frac{d}{dx} \left\{ \ln \sqrt{\frac{(1+x)e^{-2x}}{1-x}} \right\} = \frac{x^2}{1-x^2}$.

134. If $y = \frac{e^x - e^{-x}}{e^x + e^{-x}}$, show that; $\frac{d^2y}{dx^2} + 2 \frac{dy}{dx} = 0$

135. If $y = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right)$, show that; $\frac{dy}{dx} = \frac{1}{1+x^2}$.

Hence, deduce that; $\frac{d^2y}{dx^2} + 2x\left(\frac{dy}{dx}\right)^2 = 0$.

136. If $y = e^{4x} \cos 3x$, show that; $\frac{d^2y}{dx^2} - 8 \frac{dy}{dx} + 25y = 0$.

137. If $y = \frac{\sin x}{e^{-x}}$, show that; $\frac{d^2y}{dx^2} = 2 \left(\frac{dy}{dx} - y \right)$.

138. If $y = \sec x$, show that;

i). $\frac{dy}{dx} = y \tan x$

ii). $y \frac{d^2y}{dx^2} - \left(\frac{dy}{dx} \right)^2 = y^4$

139. If $y = \tan \frac{x}{2}$, show that;

i). $2 \frac{dy}{dx} = (1 + y^2)$

ii). $\frac{d^2y}{dx^2} = y \frac{dy}{dx}$

140. Given that; $y = \tan x + \sec x$, prove that;

i). $\frac{dy}{dx} = y \sec x$

ii). $\frac{d^2y}{dx^2} = y \frac{dy}{dx}$

141. If; $y = (\sin x + \cos x)^{\frac{1}{2}}$, prove that;

i). $2y \frac{dy}{dx} = \cos x - \sin x$

ii). $2y \frac{d^2y}{dx^2} + 2 \left(\frac{dy}{dx} \right)^2 + y^2 = 0$

142. If $y = \frac{x^2+1}{\sqrt{(3x+5)}}$, prove that;

i). $\frac{dy}{dx} = \frac{y(2x-3y^2)}{2(x^2+1)}$

ii). $2(x^2 + 1) \frac{d^2y}{dx^2} + (9y^2 + 2x) \frac{dy}{dx} = 2y$

143. If $y = \frac{x}{\sqrt{(x+1)}}$, prove that;

i). $\frac{dy}{dx} = \frac{y(2x-y^2)}{2x^2}$

ii). $2x^2 \frac{d^2y}{dx^2} + (2x + 3y^2) \frac{dy}{dx} = 2y$

PARAMETRIC EQUATIONS:

144. Given that; $x = (3 - 2 \cos^2 t) \cos t$, $y = (3 - 2 \sin^2 t) \sin t$, find $\frac{dy}{dx}$ at $t = \frac{\pi}{4}$

145. If $x = a \left(\frac{1+t^2}{1-t^2} \right)$, $y = \frac{2t}{1-t^2}$, find $\frac{dy}{dx}$

146. If $\sin x = \frac{2t}{1+t^2}$, $\tan y = \frac{2t}{1-t^2}$, find $\frac{dy}{dx}$

147. Differentiate $\sin^{-1} \left(\frac{2t}{1+t^2} \right)$ with respect to $\tan^{-1}(t)$

148. If $x = a \cos^3 \theta$, $y = a \sin^3 \theta$, show that; $\frac{d^2y}{dx^2} = \frac{1}{3a} \sec^4 \theta \operatorname{cosec} \theta$.

- Hence find $\frac{d^2y}{dx^2}$ at $\theta = \frac{\pi}{6}$
149. If $x = a \left(\cos t + \ln \left(\tan \frac{t}{2} \right) \right)$, $y = a \sin t$, find; $\frac{d^2y}{dx^2}$.
150. If $x = a(\theta - \sin \theta)$ and $y = a(1 + \cos \theta)$, find; $\frac{d^2y}{dx^2}$.
151. A curve is given by parametric equations;
 $x = 6t$ and $y = 1 - \cos t$
 Find; $\frac{dy}{dx}$ in terms of t .
152. A particle is moving along the path of the curve determined by the parametric equations; $x = e^{2t} + 1$ and $y = \ln(e^{4t} + 2e^{2t} + 1)$ where $t > 0$.
 i) Find $\frac{dy}{dx}$ in terms of t .
 ii) Show that; $\frac{d^2y}{dx^2} = \frac{-2}{(e^{2t}+1)^2}$.
 iii) Obtain the cartesian equation of the curve.
153. The parametric equations of the curve are; $x = e^{2t-3}$ and $y = 4 \ln t$ where $t > 0$.
 When $t = m$, the gradient of the curve is 2. Show that m satisfies the equation;
 $m = \frac{1}{2}(3 - \ln m)$
154. If $x = a \cos \theta + b \sin \theta$ and $y = a \sin \theta - b \cos \theta$, show that; $y^2 \frac{d^2y}{dx^2} - x \frac{dy}{dx} + y = 0$.
155. If $x = 3t^2 + 1$ and $y = 3t^2 + 5t$, show that; $\frac{d^2y}{dx^2} = \frac{-5}{36t^3}$
156. If $x = t^3 - t$ and $y = 3t + 6$ for $0 \leq t < 5$, show that; $\frac{d^2y}{dx^2} = -\frac{18t}{(3t^2-1)^3}$
157. If $x = 4 \sin 10t + 102$ and $y = 4 \cos 10t + 13$, find; $\frac{d^2y}{dx^2}$.
158. If $x = \log t$ and $y = \sin t$, find; $\frac{d^2y}{dx^2}$.
159. If $x = \sin \theta$ and $y = \theta - \cos \theta$, show that; $\frac{d^2y}{dx^2} = \frac{1+\sin \theta}{\cos^3 \theta}$.
160. If $x = a \cos t$ and $y = b \sin t$, show that; $\frac{d^2y}{dx^2} = -\frac{b}{a^2} \operatorname{cosec}^3 t$.
161. If $x = s^2$ and $y = s^3 - 3s$, show that; $\frac{d^2y}{dx^2} = \frac{3(s^2+1)}{4s^3}$.
162. If $x = \frac{3t-1}{t}$ and $y = \frac{t^4+1}{t}$, show that; $\frac{d^2y}{dx^2} = 12t^5$
163. If $x(t) = \frac{1}{3}t^3 - 4t$ and $y(t) = 2t^3 - 6t^2$, show that; $\frac{d^2y}{dx^2} = \frac{12}{(t^2-4)(t+2)^2}$.
164. If $x(t) = t^2 - 3t$ and $y(t) = t^3$, find; $\frac{dy}{dx}$ and $\frac{d^2y}{dx^2}$.
165. If $x(t) = t^2 - 4t$ and $y(t) = 2t^3 - 6t$, show that; $\frac{d^2y}{dx^2} = \frac{3t^2-12t+3}{2(t-2)^3}$.
166. If $x(t) = 2 \sec 5t$ and $\sqrt{3y(t)} = \tan 5t$, show that; $\frac{d^2y}{dx^2} = \frac{1}{6}$.
167. If $x(t) = t + \cos(t)$ and $y(t) = 2 - \sin(t)$, show that, at point where $t = \frac{\pi}{6}$,
 $\frac{d^2y}{dx^2} = -4$

168. If $x(t) = \sqrt{1-t}$ and $y(t) = t^3 - 3t$, show that; $\frac{d^2y}{dx^2} = 6(1 + 4t - 5t^2)$.
169. If $x(t) = (t-2)^{\frac{3}{2}}$ and $y(t) = t^2 - 1$, show that; $\frac{d^2y}{dx^2} = \frac{4(t-4)}{9(t-2)^2}$.
170. If $x(t) = \frac{1}{(t+1)}$ and $y(t) = \frac{1}{(t-1)}$, show that; $\frac{d^2y}{dx^2} = 4\left(\frac{t+1}{t-1}\right)^3$.
171. If $x(m) = 1 - \frac{1}{m}$ and $y(m) = 6 - \frac{7}{m} + \frac{2}{m^2}$, show that; $\frac{d^2y}{dx^2} = 4$.
172. If $u(x) = x^2 - 3$ and $p(x) = x^3 + x + 1$, show that; $\frac{d^2p}{du^2} = \frac{3x^2-1}{4x^3}$.
173. If $x(\beta) = (\beta^2 - 1)^2$ and $y(\beta) = 4\beta^3$, show that; $\frac{d^2y}{dx^2} = \frac{-3(\beta^2+1)}{4\beta(\beta^2-1)^3}$.
174. If $y(t) = 3t^2 - 7$ and $x(t) = 2 - 2t + t^3$, show that; $\frac{d^2y}{dx^2} = \frac{-6(3t^2+2)}{(3t^2-2)^3}$.
175. If $y(u) = u^4 - 2u^2 + \frac{4}{u}$ and $x(u) = \frac{1}{u^2}$, show that; $\frac{d^2y}{dx^2} = u^3(6u^5 - 4u^3 - 1)$.
176. If $x(t) = \ln(\cos t)$ and $y(t) = \sin(2t)$, show that;
- $\frac{dy}{dx} = -2 \cos(2t) \cot(t)$
 - $\frac{d^2y}{dx^2} = \frac{-2 \cos t (\sin^2(2t) + \cos(2t))}{\sin^3 t}$
177. If $x(t) = e^t \cos(t)$ and $y(t) = \ln(1 + t^2)$, show that; $\frac{dy}{dx} = \frac{2te^{-t}}{(1+t^2)(\cos t - \sin t)}$.
178. If $y(t) = \tan(t)$ and $x(t) = \ln(1 + \sin t)$, show that;
- $\frac{dy}{dx} = \sec^3 t (1 + \sin t)$
 - $\frac{d^2y}{dx^2} = \sec^5 t (1 + \sin t)^2 (1 + 2 \sin t)$
179. Given that $y e^x + \ln y = 0$. Show that; $\frac{dy}{dx} = \frac{\ln(y^y)}{1 - \ln y}$.
- Hence, find the value of $\frac{dy}{dx}$ at a point where $y = 0.3$.

SMALL CHANGES:

180. Using small change, show that; $\sqrt{1.98} \cong 1 \frac{57}{140}$.
181. Using small changes, show that; $\sqrt{1.42} \cong 1 \frac{23}{120}$.
182. Using small changes, show that; $\sqrt[3]{215} \cong 5 \frac{107}{108}$.
183. Using small changes, show that; $\sqrt[3]{29.0} \cong 3 \frac{2}{27}$.
184. Using small changes, show that; $\sqrt[3]{515.0} \cong 8 \frac{1}{64}$.
185. Using small changes, show that; $\sqrt{1.7} \cong 1 \frac{79}{260}$.

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186. Using small changes, show that; $(627.5)^{\frac{1}{4}} \cong 5 \frac{1}{200}$.
187. Using small changes, show that; $\sqrt[4]{81.12} \cong 3 \frac{1}{900}$.
188. Using small changes, show that; $\sqrt[4]{31.5} \cong 1 \frac{159}{160}$.
189. Using small changes, show that; $\sqrt[6]{730} \cong 3 \frac{1}{1458}$.
190. Using small changes, show that; $\sin(30.2^{\circ}) \cong \frac{1}{1800} (900 + \pi\sqrt{3})$.
191. Using small changes, show that; $\sin(29.6^{\circ}) \cong \frac{1}{900} (450 - \pi\sqrt{3})$.
192. Using small changes, show that; $\tan(60.4^{\circ}) \cong \frac{1}{225} (225\sqrt{3} + 2\pi)$.
193. Using small changes, show that; $\tan(44.8^{\circ}) \cong \frac{1}{450} (450 - \pi)$.
194. Using small changes, show that; $\cos(30.5^{\circ}) \cong \frac{1}{720} (360\sqrt{3} - \pi)$.
195. Using small changes, show that; $\cos(28.6^{\circ}) \cong \frac{1}{1800} (900\sqrt{3} + 7\pi)$.
196. Using small changes, show that; $\sec^2(29.7^{\circ}) \cong \frac{1}{675} (900 - \pi\sqrt{3})$.
197. Using small changes, show that; $\sin^2(30.2^{\circ}) \cong \frac{1}{1800} (450 + \pi\sqrt{3})$.
198. Using small changes, show that; $\cot^2(45.3^{\circ}) \cong \frac{1}{150} (150 - \pi)$.
199. Using small changes, show that; $\ln(3.5) \cong -\frac{1}{8} + 2\ln 2$.
200. Using small changes, show that; $\ln 2.1 \cong \frac{1}{20} + \ln 2$.
201. Using small changes, show that; $\ln 10.0 \cong \frac{1}{9} + 2\ln 3$.
202. Using small changes, show that; $\ln 3.8 \cong -\frac{1}{20} + 2\ln 2$.
203. Using small changes, show that; $\ln 8.4 \cong \frac{1}{20} + 3\ln 2$.
204. A 3% error was made in measuring the area of the hemisphere, find the corresponding percentage error in the;
a) Surface area,
b) Volume, of the same sphere.
205. An error of 6% is made in measuring the volume of the sphere. Find the corresponding percentage error made in measuring the surface area of the same sphere.
206. The radius of a sphere is measured to be 5.0cm with a possible measurement error of 0.1cm . Calculate the corresponding percentage error in the volume of the same sphere.
207. The side of a square is measured as 10.0cm . If there is a percentage error of 2% in measuring the side of the square, find the resulting percentage error in its area.
208. The side of a square is 5.0cm . Find the increase in the area of the square when the

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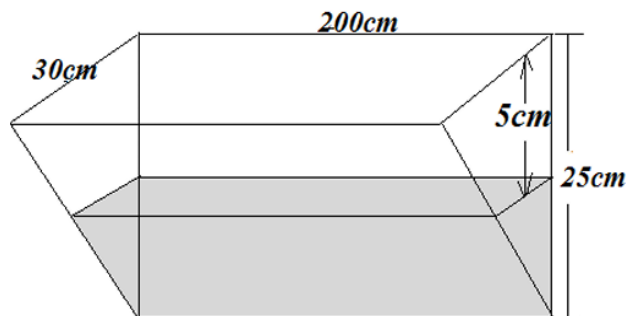
- side increases by 0.01cm .
209. A 5% is made in measuring the radius of a sphere.
Find the corresponding percentage error in the surface area of the same sphere.
210. An error of $2\frac{1}{2}\%$ is made in the measurement of the area of the circle.
What corresponding percentage error results in the;
i) Radius,
ii) Circumference, ?
211. The base radius of a circular cone increases and the volume changes by 2%. If the height of the cone remains constant, find the percentage increase in the circumference of the base.
212. The period T of a simple pendulum is given by; $T = 2\pi\sqrt{\frac{l}{g}}$ where l is the length of the pendulum and g is the acceleration due to gravity.
If the length of the pendulum, l is increased by 4%, what is the corresponding percentage increase in the period, T of the pendulum?
213. The radius of a sphere is measured with percentage error of 1.5%.
Calculate the corresponding percentage error in its surface area.
214. The side of a cube is measured to be 10cm . If there is an error of 2% in measuring its side, calculate the corresponding percentage error in the calculated volume of the same cube.
215. One side of a rectangle is three times the other. If its perimeter increases by 2%, what's the corresponding percentage increase in its area?
216. The height of a closed cylinder is thrice its radius. Find the percentage increase in its total surface area corresponding to a unit percentage increase in height.
217. The volume of a sphere increases by 2%.
Find the corresponding percentage increase in its surface area.
218. The volume of a sphere is increased by 3%.
Find the corresponding percentage increase in its radius.
219. The height, h , of a right angled – triangle increases at a constant rate of 0.05m per hour while keeping the base constant at 5 metres . Find the rate of the change of the angle, θ , opposite to the height, h , when $\theta = \frac{\pi}{3}$.

RATES OF CHANGE:

220. The side of a square is increasing at a rate of 5cms^{-1} . Find the rate of increase of the area when the length of the side is 10cm .

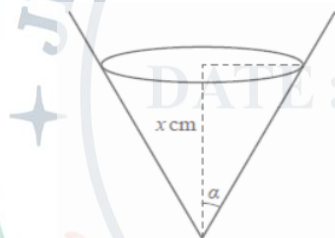
221. A spherical balloon is being inflated such that its volume is increasing at a rate of $50\text{cm}^3/\text{s}$. How fast is its radius increasing when the diameter is 20cm ?
222. A conical water container with its vertex down has a height of 10cm and a radius of 5cm . Water is being poured into the container at a rate of $2\text{m}^3/\text{min}$. Find the rate at which the water level is rising when the water is half way up the container.
223. The area of a circle is increasing at a rate of $4\pi\text{cm}^2\text{s}^{-1}$. Find the rate of increase of the length of side when the volume is 125cm^3 .
224. The area of a square of side $x\text{cm}$ is increasing at a rate of $6\text{cm}^2\text{s}^{-1}$. Show that the expression of the rate of increase of the length of the side is $\frac{3}{x}\text{cm/s}$.
225. A boy is inflating a spherical balloon at a rate of $10\text{cm}^3\text{s}^{-1}$. Find the rate of increase of surface area of the balloon when the radius is 5cm .
226. A container in the shape of a hollow cone of semi – vertical angle 45° is held with its vertex pointing downwards. Water drips into the container at a rate of 3cm^3 per minute. Find the rate at which the depth of water in the cone is increasing when the depth is 2cm .
227. Sand pouring at a constant rate of $\left(\frac{1000}{3}\pi\right)\text{cm}^3\text{s}^{-1}$ forms a right circular conical heap of height, **10cm**. Express the volume of the heap in terms of the vertical angle. Hence find the rate of change of the vertical angle when its 60° .
228. An inverted cone with a vertical angle 60° has water in it dripping out through the hole at the vertex at a rate of $3\text{cm}^3/\text{min}$. Find the rate at which the surface area of the cone in contact with water is changing at the instant when the volume left in the cone is $8\pi\text{cm}^3$.
229. Water runs at a constant rate of $6\text{cm}^2\text{s}^{-1}$ in the vessel whose volume is obtained by rotating the area bounded by the curve; $4y = x^2$ about the $y - \text{axis}$ from $y = 0$ to $y = h\text{cm}$.
 - i) Show that the volume of the vessel is $2\pi h^2\text{cm}^3$
 - ii) Find the rate at which the water level is rising when the water has been running for 3 seconds.
230. The side of a square is increasing at the rate of 5m/s . Find the rate of increase of the area when the length of the side is 10cm .
231. A container in the shape of a right circular cone of height 10cm and base radius 1.0cm is catching the drips from a tap leaking at a rate of $0.1\text{cm}^3\text{s}^{-1}$. Find the rate at which the surface area of water is increasing when the water is halfway up the cone.
232. A horse trough has a triangular cross – section of height 25cm , base 30cm and 2m long. A horse is drinking steadily and when the water is 5.0cm below the top, it is being lowered at a rate of 1.0cm per minute. Find the rate of consumption in litres per minute.

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Ans: 4.8litres/min

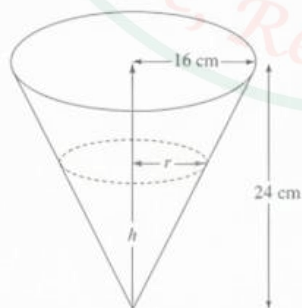
233. An inverted circular cone with radius 3cm and a height 9.0cm is initially full of water.
Water is leaking from the circular base of the cone at a constant rate of $2.0\text{cm}^3\text{s}^{-1}$. Find the exact rate of change of the depth of water when the depth of water is 6cm .
234. An ice cube is melting such that the rate of decrease of the sides of the cube is 5.0mm/min . Find the rate of decrease of the surface area of the ice cube when the side length is 2.0cm .
235. The volume of a spherical balloon is increasing at a rate of $0.05\text{m}^3/\text{s}$. How fast is the surface area increasing when the radius is 0.4m ?
- 236.



The diagram above shows a container in the form of a right circular cone. The angle between the axis and the slant height is α where $\alpha = \tan^{-1}\left(\frac{1}{2}\right)$. Initially the container is empty, and then the liquid is added at a rate of $14\text{cm}^3/\text{min}$. The depth of liquid in the container at a time, t minutes is $x\text{cm}$.

- Show that the volume, $V\text{cm}^3$, of the liquid in the container when the depth is $x\text{cm}$ is given by; $V = \frac{1}{12}\pi x^3$
 - Find the exact rate at which the depth of the liquid is increasing at the instant where the depth is 8.0cm .
237. A cylinder of radius r and height $5r$. The radius increases from 4cm to 4.1cm . Find the approximate increase in volume.
238. A container in shape of an inverted right circular cone, the height is 12cm and with a vertical angle of 60° . A tap delivers water in the container at a rate which is proportional to the depth of water collected in the container at any time, t .

- i) Show that the rate at which the water level is rising is inversely proportional to the depth.
 - ii) If it takes 1 minute to collect a depth of 6cm of water, calculate the time it takes to fill the whole container.
239. A conical vessel whose height is 16cm and the base radius is 8cm is being filled by water at a uniform rate of $1.5\text{m}^3/\text{s}$. Find the rate at which the level water in the vessel is rising when the depth is 4cm .
240. A vessel is in the shape of an inverted right circular cone. The radius of the top is 8.0cm and the height is 40.0cm . Water is poured into the vessel at a rate of $20\text{cm}^3/\text{s}$. Calculate the rate at which the water level is rising when;
- i). The water level is 12cm from the top.
 - ii). The vessel is one quarter full.
241. Water is leaking from the conical funnel at a rate of $5\text{cm}^3/\text{s}$. If the radius of the base of the funnel is 5cm and its altitude is 10cm , find the exact rate at which the water level is dropping when it is 2.5cm from the top.
242. A conical funnel 12cm in diameter and 24cm deep is filled with a liquid that runs out at a rate of $5\text{cm}^3/\text{min}$. At what rate is the;
- i). Depth of the liquid change when its depth is 10.0cm ?
 - ii). Radius of the surface of water changing when the water level is 8.0cm from the top?
243. Water is dripping through a tinny hole at the vertex in the bottom of the conical funnel at a uniform rate of $4\text{cm}^3/\text{s}$. When the slant height of the water is 3.0cm , find the rate of decrease of the slant height of water, given that the vertical angle of the funnel is 120° .
- 244.



A container is made in the shape of a hollow inverted right circular cone. The height of the container is 24.0cm and the radius is 16.0cm as shown above. Water is

flowing into the container. When the depth of water is h cm, the surface of water has radius r cm and the volume of water is V cm³.

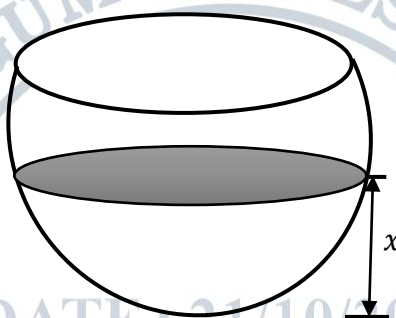
i). Show that $V = \frac{4\pi h^3}{27}$.

Water flows into the container at a rate of 8 cm³/s.

ii). Find, in terms of π , the rate at which the water level is rising when the water level is at 10.0 cm mark from the top of the container.

iii). Find the exact rate at which the radius of the surface of water is increasing at an instant when the water level is three quarters way up the top of the container.

245.



Water is poured into a hemispherical bowl of radius 6.0 cm at a rate of 20 cm³/s. After t seconds, the volume of water in the bowl, V cm³, is given by;

$$V = 6\pi x^2 - \frac{1}{3}\pi x^3$$

Where x cm is the depth of water in the bowl.

Calculate the exact rate at which the water level in the bowl is rising when the depth of water is 4.0 cm.

246. Water runs into a conical vessel fixed with its vertex downwards at a rate of 3 cm³/s filling the vessel to a depth of 15.0 cm in a time of one minute. Find the rate at which the depth of water in the vessel has been running for 7.5 seconds.

247. A hemispherical bowl is being filled with water at a uniform rate. When the height of water in the bowl is h cm, the volume, V cm³, is;

$$V = \pi \left(rh^2 - \frac{1}{3}h^3 \right) \text{ cm}^3, r \text{ cm being the radius of the hemisphere.}$$

Calculate the rate at which the Water level is rising when its half way to the top, given that $r = 6.0$ cm and that the bowl fills in 1 minute.

248. A hemispherical bowl of radius, a cm is initially fully of water.

The water runs out of a small hole of the bowl at a constant rate which is such that it will empty the bowl in 24s.

Given that when the depth of water is $x\text{cm}$, the volume of water in the bowl is;

$$V = \frac{1}{3}\pi x^2(3a - x)\text{cm}^3.$$

i). Prove that the depth is decreasing at a rate of; $\frac{a^3}{36x(2a-x)}\text{cms}^{-1}$

ii). Find after what time the depth of water is; $\frac{1}{2}a\text{cm}$, and the rate at which the water level is then decreasing? **Ans:** $13\frac{1}{2}\text{s}$

249. An inverted cone has a diameter of 28.0cm at the top and a height of 24.0cm . At the instant when the water in the container is 0.16m deep, the surface level is falling at a rate of $8\text{cm}/\text{min}$. Find the rate at which water is being drained.
250. A conical tank has a vertical angle of 60° . Water flows through a hole at the vertex of the tank at a rate of 6cm^3 in one second. At what rate is the water level falling when the volume of water in the cone is $24\pi\text{cm}^3$.
251. An ice cream cone has a radius of 5.0cm and height of 15.0cm . Ice cream melts and drips out of the cone vertex at a rate of $0.01\text{m}^3/\text{s}$. Find the rate at which;
- height,
 - radius, of the ice cream decreases when the radius is 2cm .
252. A cylindrical tank with radius 5m is being filled with water at a rate of $3\text{m}^3/\text{min}$. How fast is the height of water increasing?

TANGENTS AND NORMALS:

253. The curve, $f(x)$ has the equation; $f(x) = 9 - 4x - \frac{8}{x}$, $x > 0$. The point P on the curve has x -coordinate equal to 2.
- Show that the equation of the tangent to $f(x)$ at the point P is; $y = 1 - 2x$.
 - Find the equation of the normal to $f(x)$ at point P
- The tangent at P meets the x -axis at A and the normal at P meets the x -axis at B.
- Find the coordinates of A and B. Hence, calculate the area of the triangle APB.
254. The curve, $f(x)$ has the equation. $f(x) = \frac{(x+3)(x-8)}{x}$.
- Find $\frac{dy}{dx}$ in its simplest form.

- b) Find the equation of the tangent to $f(x)$ at the point where $x = 2$.
255. The curve, $f(x)$ has the equation; $f(x) = \frac{1}{2}x^3 - 9x^{\frac{3}{2}} + \frac{8}{x} + 30$, $x > 0$.
- a) Find $\frac{dy}{dx}$
- b) Show that the point $P(4, -8)$ lies on $f(x)$
- c) Find the equation of the normal to $f(x)$ at point P, giving your answer in the form; $ax + by + c = 0$.
256. The curve has the equation; $y = x^3 - 2x^2 - x + 9$.
The point M has coordinates (2,7)
- a) Show that M lies on the curve.
- b) Find the equation of the tangent to the curve at M, giving your answers in the form; $y = mx + c$, where m and c are constants.
- The point Q also lies on the curve.
- Given that the tangent to the curve at Q is perpendicular to the tangent to the same curve at M.
- c) Show that the x -coordinate of Q is; $\frac{1}{3}(2 \pm \sqrt{6})$
257. Find all the points on the curve $y = (x + 3)^3$ at which the tangent line is parallel to the line with equation; $y - 4x + 2 = 0$.
258. Find the exact equation of the tangent line to the curve given by; $\sin(x + y) = 2x - 2y$ at the point; (π, π) .
259. Find the equation of the tangent to the curve; $(x^2 + y^2)^2 = \frac{25}{3}(x^2 - y^2)$ at the point (2,-1).
260. Find the equation of the tangent and normal to; $f(x) = [\sin^{-1}(x)]^2$ at the point where; $x = \frac{-1}{\sqrt{2}}$.
261. Find the equation of the tangent to the curve; $f(x) = e^{\tan^{-1}(x)}$ at the point where the curve cuts x - axis.
262. Find the equation of the normal to the curve; $y = \tan^{-1}(5x)$ at $(\frac{1}{5}, \frac{\pi}{4})$.
263. The curve, $f(x)$ has the equation; $x = 8y \tan 2y$.
The point P has the coordinates; $(\pi, \frac{\pi}{8})$.
- a) Verify that P lies on $f(x)$
- b) Find the equation of the tangent to $f(x)$ at P in the form; $ay = x + b$, where the constants; a and b are in terms of π .

264. The point $N\left(x, \frac{\pi}{4}\right)$ is the point on the curve; $x = 2 \tan\left(y + \frac{\pi}{12}\right)$.
Find the equation of the normal to the curve at point N.
265. A curve is given parametrically by the equations.
 $x = 2t^2 - 3t + 1$, $y = t^2 + t + 1$
The tangents to the curve, at two distinct points P and Q, intersect each other at point; (2,9).
a) Determine the coordinates of P and Q
b) Show that the cartesian equation of the curve is given by;
 $25(y - 1) = (2y - x - 1)(2y - x + 4)$.
266. A curve, $f(x)$ is given by the parametric equations;
 $x = t^2 - t$ and $y = \frac{at}{1-t}$
a) Find the cartesian equation of the tangent to $f(x)$ at the point where $t = -1$.
b) Determine the coordinate of the point where the above tangent meets $f(x)$ again.
267. A curve, $f(x)$ is parametrically given by; $x = 1 - 3t$ and $y = \frac{6+t}{2+t}$.
a) Show that the line; $l: 4x - 3y = 1$ is a tangent to the curve, $f(x)$.
b) At what point(s) does the line, l above intersect $f(x)$.
268. The point P(8,9) lies on the curve, $f(x)$ with parametric equations;
 $x = 2t^2$ and $y = 3^t$
The tangent to the curve, $f(x)$ at P meets the y -axis at point Q.
Determine the exact coordinates of Q. **Ans: Q** $\left(0, 9 \ln\left(\frac{e}{3}\right)\right)$
269. A curve, $f(x)$ is parametrically given by;
 $x = \frac{1}{t} + \frac{1}{t^2}$ and $y = \frac{1}{t} - \frac{1}{t^2}$, $t \neq 0$.
Show clearly that;
i). $\frac{dy}{dx} = \frac{(t-2)}{(t+2)}$
ii). the equation of the tangent to $f(x)$ at the point where $t = \frac{1}{2}$ is;
 $3x + 5y = 8$
iii). the cartesian equation of the curve is; $\frac{(x+y)^2}{(x-y)} = 2$
270. The curve, $f(x)$ has parametric equations;
 $x = \sin 2\theta$ and $y = 2 \cos^2 \theta$, $0 \leq \theta \leq 2\pi$.
a) Show clearly that; $\frac{dy}{dx} = -\tan 2\theta$

- b) Find the exact equation of the tangent to $f(x)$ at the point where; $\theta = \frac{\pi}{3}$.
- c) Show that the cartesian equation of the curve, $f(x)$ is given by;
- $$x^2 = y(2 - y)$$
271. A curve, $f(x)$ is given by the parametric equations;
 $x = 2\cos\theta + \sin 2\theta$ and $y = \cos\theta - 2\sin 2\theta$, $0 \leq \theta \leq 2\pi$.
 The point P lies on $f(x)$ where $\theta = \frac{\pi}{2}$.
- a) Find the coordinates of point P
- b) Show that the equation of the normal to $f(x)$ at point P is given by;
- $$4x + 2y = 5\sqrt{2}$$
272. A curve, $f(x)$ is given by the parametric equations;
 $x = \sec\theta$, $y = \ln(1 + \cos 2\theta)$; $0 \leq \theta \leq \frac{\pi}{2}$
- a) Show clearly that; $\frac{dy}{dx} = -2\cos\theta$
- b) Show that the cartesian equation of the;
- line, l that's a tangent to $f(x)$ can be written as; $x + y = 2 - \ln 2$.
 - curve, $f(x)$ is given by; $x^2 e^y = 2$.
273. The tangent to the curve, $x^2 + xy + 2y^2 = 7$ at a point M is parallel to the x -axis. Find the possible coordinates of point M.
274. A curve, $f(x)$ is defined by the following parametric equations.
- $$x = \frac{t}{1+t^2} \text{ and } y = \frac{2t^2}{1+t^2}$$
- The straight line, $l: y + 2 = 6x$ intersects the curve; $f(x)$ at points R and T.
- a) Show that; $\frac{dy}{dx} = \frac{4t}{1-t^2}$.
- b) Find the exact coordinates of points R and T.
275. A curve, $f(x)$ is given by the parametric equations;
- $$x = \frac{3t-2}{t-1} \text{ and } y = \frac{t^2-2t+2}{t-1}$$
- The point P $\left(1, -\frac{5}{2}\right)$ lies on $f(x)$.
 Show clearly that;
- a) $\frac{dy}{dx} = t(2 - t)$
- b) the equation of the tangent to $f(x)$ at the point P is; $3x = 4y + 13$.
276. A curve, $f(x)$ is parametrically given by the equations;

$$x = t^2 - 8t + 12 \text{ and } y = t - 4$$

The point P $(-3,1)$ lies on the curve, $f(x)$.

- a) Find the coordinates of the points where $f(x)$ cuts the coordinate axes.
 - b) Show clearly that;
 - i). the equation of the normal to $f(x)$ at P is; $y + 2x + 5 = 0$
 - ii). cartesian equation of $f(x)$ is; $y^2 = x + 4$.
277. Find the equations to the tangent lines of the curve $y = (x + 8)(x + 10)$ at the points where this curve intersects x -axis.
278. Find the points where the tangent is parallel to; x -axis and y -axis.
- a) $x^3 + y^3 = 3axy$
 - b) $25x^2 + 12xy + 4y^2 = 1$.
279. Find all the points on the given curve where the tangent line is
- a) horizontal b) vertical
 - i). $x^2 - xy + y^2 = 3$
 - ii). $x^2 + xy + y^2 = 3$
280. Find the equations of the two lines through the origin that are tangent to the curve; $x^2 - 4x + y^2 + 3 = 0$
281. A curve through A(5,1) has the equation; $y = 3 - \frac{10}{x}$.
- a) Show that the cartesian equation of the normal to the curve at A is; $5x + 2y = 27$
- The normal meets the curve again at M.
- b) Find the coordinates of point M.
 - c) Find the midpoint of AM.
282. A curve has equation; $y = 3x - \frac{4}{x}$ and passes through the points A(1,-1) and B(4,11). At each of the points C and D on the curve, the tangent is parallel to AB. Find the equation of the perpendicular bisector of CD.
283. The tangents to the curve; $y = 5 - 3x - x^2$ at the points; $(-1,7)$ and $(-4,1)$ intersect at point R. Find the coordinates of point R.
284. The normal to the curve; $y = 4 - 2\sqrt{x}$ at point P(16,-4) meets the x -axis at point N.
- a) Find the cartesian equation of the normal.
 - b) Find the coordinates of N.
285. The equation of the curve; $y = 2x - \frac{10}{x^2} + 8$.

- a) Find $\frac{dy}{dx}$
- b) Show that the normal to the curve at the point $\left(-4, -\frac{5}{8}\right)$ meets the y -axis at point $(0, -2.995)$
286. The normal to the curve; $y = \frac{6}{\sqrt{x-2}}$ at the point $(3, 6)$ meets the x -axis at P and the y -axis at Q.
Find the midpoint of PQ
287. A curve has equation; $y = x^5 - 8x^3 + 16x$. The normal at the point S $(1, 9)$ and the tangent at the point B $(-1, -9)$ intersect at point D.
Find the coordinates of point D.

MAXIMA, MINIMA AND OPTIMIZATION:

288. Find and distinguish the exact stationary (turning) points of each of the following curves.
- $y = x^3 + 6x^2 + 9x + 4$
 - $f(x) = 2x^3 - 3x^2 - 72x + 5$
 - $y = 2x^3 - 3x^2 - 36x + 10$
 - $f(x) = (x^2 - 1)(x + 2)$
 - $y = 2x(x - 5)(3x + 2)(4x - 3)$
 - $f(x) = 4 - 5x + 4x^2 - 3x^3$
 - $y = -4x^4 + 3x^2 - 15x + 5$
 - $y = 2x^4 - 8x^3 + 9x^2$
 - $h(x) = 2x^3 - 3x^2 - 12x + 5$
289. Obtain the exact stationary point(s) of each of the following implicit curves. Distinguish their nature.
- $xy + y^2 = x^2 + 5$
 - $x^2 + 2xy - 3y^2 = 4x + 4y - 20$
 - $x^2 + 4xy + 2y^2 + 18 = 0$
 - $2x^2 + xy - y^2 - 4x - y + 20 = 0$
 - $\frac{3x^2}{y} - 5y = 2(x + 8)$
 - $xy(x - y) = -16$
 - $\sin 2x \cot y = 1$
 - $x^2 - 2y^2 - xy - x + 5y + 34 = 0$

290. Determine the exact stationary points of each of these curves given their parametric coordinates. Distinguish them.
- $x = t + \ln t$ and $y = t - \ln t$
 - $x = 1 - \cos 2t$ and $y = \sin 2t$
 - $x = \frac{t}{1-t}$ and $y = \frac{t^2}{1-t}$
 - $y = 2t + \frac{1}{2t} + 2$ and $x = 2t - \frac{1}{2t}$
 - $x = \sin^2 \theta$ and $y = \sin 2\theta$
 - $x = 2t + 1$ and $y = 8t^3 + 4t^2$
 - $x = t^4$ and $y = 2t^2 - 8t + 9$
 - $x = 4\theta - \cos \theta$ and $y = 1 + \sin \theta$.
291. An open box with a square base has a surface area of $48.0m^2$. Find the dimensions of the same box that maximizes its volume. Hence, calculate its maximum volume.
292. A rectangular field of $500m^2$ is to be fenced using a mesh. One side of the field is a straight river. This side is not to be fenced. Find the dimensions of the field that will minimize the wire mesh to be used.
293. A rectangular figure with sides $8cm$ by $5cm$, equal sides are removed from each corner and the edges are turned up to make an open box of volume Vcm^3 . Show that; $V = 40x - 26x^2 + 4x^3$, and hence, find the maximum possible volume and the corresponding value of x .
294. The sum of the height and radius of the right circular cone is $9.0cm$. Show that the maximum volume of the cone will be $36\pi cm^3$.
295. The length of a rectangular block is three times its width. The total surface area of the block is $180.0m^2$. Calculate its maximum volume.
296. Determine the largest area of the rectangular piece of land that can be fenced by $100m$ piece of a fencing wire if the part of the already existing wall on the longer side of the land is to be used.
297. Show that the maximum volume, V of cylinder that can be inscribed in a sphere of radius, R is given by; $V = \frac{4\pi R^3}{3\sqrt{3}}$ cubic units.
298. A right circular cone of vertical angle 2θ is inscribed in a sphere of radius R . Show that the area of the curved surface of the cone is $\pi R^2(\sin 3\theta + \sin \theta)$. Hence, prove that its greatest area is; $\frac{8\pi R^2}{3\sqrt{3}}$ Square units.
299. A piece of a wire of length l is cut into two portions; x and $l - x$.

Each piece is cut into twelve equal parts soldered together so as to form the edges of the cube. Show that the total volume of the cubes so formed is given by;

$$V = \frac{(l^3 - 3xl^2 + 3lx)}{1728}, \text{ and that its minimum volume is; } \frac{l^3}{6912}$$

300. A printed page must contain 60cm^2 of printed material.

There are to be margins of 5.0cm on either side and margins of 3.0cm on the top and bottom.

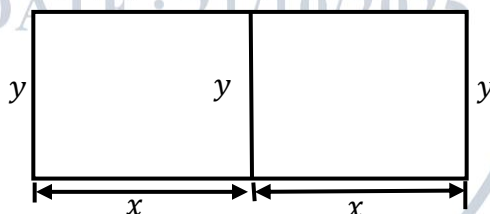
Calculate the dimensions of the printed area that can minimize the amount of paper to be used.

301. A rectangular box with open top is to be formed from a rectangular piece of cardboard which is 8cm by 3cm . Find the size square that should be cut from each corner of the cardboard to form the box with maximum volume.

302. A cylinder is inscribed in a cone that has a height of 5.0cm and radius of 2.0cm . Find the maximum surface area of the cylinder that can be inscribed in the cone.

303. A pen is to be constructed with 200.0m of fence.

It must consist of two congruent rectangular paddocks with one common side. Determine the dimensions that maximize the pen's area. Hence, calculate the maximum area of the pen.



304. If 2700cm^2 of the material is available to make a box with a square base and open top, find the dimensions a box should have to have the maximum possible volume.

305. The length of a rectangular box is 3 times its width. If the volume is 972cm^3 , find the dimensions of the box if the surface area is to be minimum.

306. A box with an open top can be made from a square sheet of cardboard of side 12.0cm , by cutting out a square of side $x\text{cm}$ from each corner, and folding it up. Find the maximum volume of the box so formed.

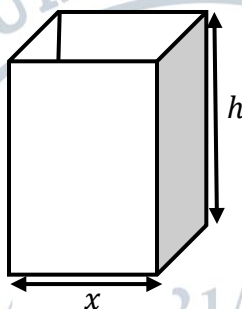
307. A tank is in the shape of a closed right circular cylinder of radius $r\text{ cm}$ and height $h\text{ cm}$. The tank has a volume of $16\pi\text{m}^3$ and is made of thin sheet metal. Find the height and radius of the same tank if its surface area is minimum.

308. In a factory, the products are always packed in rectangular boxes.

A box is being designed with a cover top such that it has a total surface area of 720cm^2 . Given that its length is to be 5 times its width, x ;

- i) Show that the volume of the box, $V \text{ cm}^3$, is given by; $V = 300x - \frac{25}{6}x^3$
- ii) Find the dimensions of the box if it's to have maximum volume.
- iii) Show that the maximum volume of the box is; $V_{\max} = 400\sqrt{6} \text{ cm}^3$.
309. A rectangular box whose width, x is a fourth of its length has a total surface area of 1728 cm^2 and no top cover. Given that its height is h ;
- i) Show clearly that; $h = \frac{(864-2x^2)}{5x}$
- ii) Show that the volume of the box is; $V = \frac{8}{5}(432x - x^3)$
- iii) Show that the maximum volume of the box is **5529.6 cm^3** .

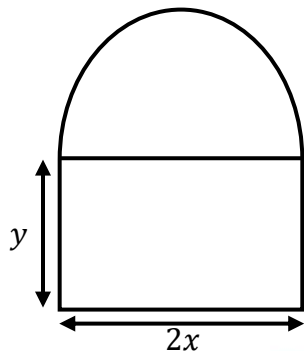
310.



The figure above shows the designs of an open large water tank in the shape of a cuboid with a square base. Given that the capacity of the tank is 500 m^3 ,

- i) Clearly, show that the surface area of the tank is; $A = x^2 + \frac{2 \times 10^3}{x}$.
- ii) Hence, show that; $A_{\min} = 300 \text{ m}^2$.
- iii) Find the dimensions of the tank for the tank to have minimum surface area.
311. A cylindrical can closed at both ends of radius, $r \text{ cm}$ and height, $h \text{ cm}$ has a surface area of $192\pi \text{ cm}^2$.
Show that the maximum volume of the can is; **$256\pi\sqrt{2} \text{ cm}^3$** . And find the exact dimensions of the can for this to occur.
312. A thin – right circular cylindrical tank closed at both ends has a capacity of $16\pi \text{ cm}^3$.
- i) Show that the least possible surface area of the tank is; $24\pi \text{ cm}^2$.
- ii) Determine the dimensions of the tank to have least possible surface area.

313.



The figure above shows the design of a window which is in the shape of a semi-circle attached to a rectangular frame. The total perimeter of the window is $6m$.

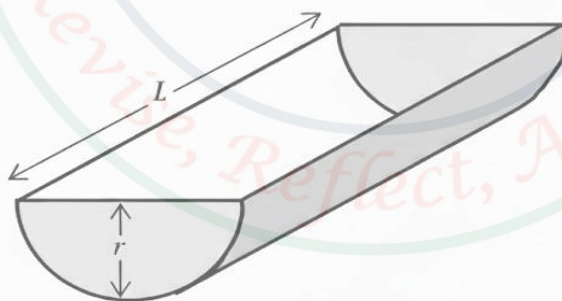
- i) Show that the total area of the window, A is given by;

$$A = 6x - \frac{1}{2}(4 + \pi)x^2$$

- ii) Show that the greatest possible area of the window is;

$$\frac{18}{(4 + \pi)} m^2$$

314. The figure below shows the design of an animal feeder which is in the shape of a hollow, open topped half cylinder, made of thin sheet metal. The radius of the semi-circular ends is r cm and the length of the feed is l cm. The metal used to form the feeder is 600π cm².

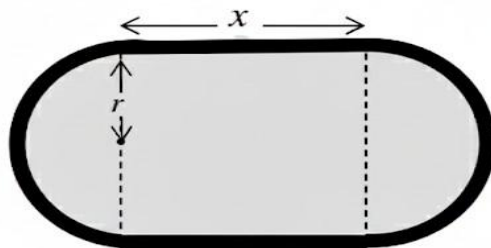


- i) Show that the capacity, V cm³ of the feeder is given by;

$$V = 300\pi r - \frac{1}{2}\pi r^3$$

- ii) Find the exact dimensions of the feeder if the feeder is to have the greatest possible capacity.
iii) Hence, show that; $V_{max} = 2,000\pi\sqrt{2}$ cm³.

315.



The figure above shows the design of an Athletics track inside a stadium. The track consists of two semi – circles, each of radius, r m, joined up to a rectangular section of length, x metres. The track is 400m all round, and has an area, A .

i) Show that; $A = 400r - \pi r^2$.

ii) Clearly, show that the maximum area of the track occurs when; $r = \frac{200}{\pi}$ m

iii) Hence, show that; $A_{max} = \left(\frac{40,000}{\pi}\right) m^2$.

316. An open box is to be made of thin sheet metal, in the shape of a cuboid with a square base of length x m and height h m. The box has a fixed volume.

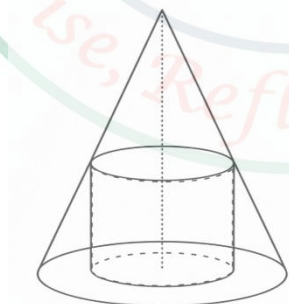
Clearly, show that the box has the least possible surface area when $x = 2h$.

317. A solid right circular cylinder of fixed volume has radius r and height h .

Show, clearly, that when the surface area of the cylinder is minimum, then;

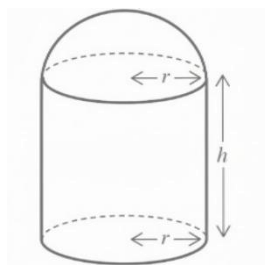
$h:r = 2:1$

318. A solid right circular cylinder is to be cut out of a solid right circular cone whose radius is 1.5m and its height is 300cm as shown below.



Show that the greatest possible volume of the cylinder to be cut out is; πm^3 .

319.



The figure above shows a hollow container consisting of a right circular cylinder of radius $r \text{ cm}$ and of height $h \text{ cm}$ joined to a hemisphere of the same radius.

The cylinder is open on one end of the circular ends and the hemisphere is also open on its circular base. The cylinder is joined to the hemisphere at their open ends so that the resulting object is open at only one end.

Given that the volume of the resulting object is; $2880\pi \text{ cm}^3$,

i) Show that, the total surface area of the container, $S \text{ cm}^2$ is given by;

$$S = \frac{5\pi}{3r} (r^3 + 3456)$$

ii) Show further that the resulting object has the least possible surface area when; $r = h$. Hence, calculate the exact least surface area.

320. A rope of length $l \text{ cm}$ is to be cut into two portions. The first portion is folded into a square of side $x \text{ cm}$, and the remaining portion is folded into a rectangle whose its width is equal to the side of the square formed from the first piece.

i). Show that the length of the rectangle is; $\frac{1}{2}(l - 6x) \text{ cm}$

ii). If the sum of the areas of the square and that of the triangle is to be maximum, show that the area of the square is; $\frac{1}{64}l^2 \text{ cm}^2$

CURVE SKETCHING 1 AND AREA UNDER THE CURVE:

321. Sketch the following curves, and in each case, find the area enclosed between the curve and x -axis.

i). $y = x(x - 1)(x - 3)$

ii). $y = 3x(x^2 - 1)$

iii). $y = -x^3 + 7x^2 - 10x$

iv). $y = x(x - 1)(x - 5)$

v). $y = (x + 1)(x - 5)$

vi). $y = x(x^2 - x - 2)$

vii). $y = -\frac{1}{4}x(x+2)(x-1)(x-4)$

322. On the same axes, sketch the following two curves, and in each case, calculate the exact area enclosed between two curves.

i). $y = 2x^2 - 6$ and $y = 10 - 2x^2$.

ii). $y = x^2$ and $y = 2x^2 - 25$.

iii). $y = x^2(1+x)$ and $y = 2x(1+x)$

iv). $y = x^2(x^2 - 2)$ and $y = 2x^2$

v). $y = x^3 - 4x^2 + 3x + 1$ and $y = x^2 - 3x + 1$

vi). $f(x) = x^2$ and $g(x) = x(2-x)$

vii). $y = x^2 - 7x + 9$ and $y = -x^2 + 3x + 1$

viii). $y = x^3 - 6x + 3$ and $y = x^2 + 3$

ix). $y = x^2 - 3(x-1)$ and $y = x^3 - 3x^2 + 3$

x). $f(x) = 3x^3 - x^2 - 10x$ and $g(x) = -x(x-2)$

xi). $g(x) = x^2$ and $f(x) = x^3$

xii). $y = 3\sin x$ and $y = -2\cos 2x$ for $0 \leq x \leq 2\pi$

xiii). $g(x) = x^3 - 2x^2 - x + 2$ and $f(x) = x^2 - 3x + 2$.

xiv). $y = \sqrt{x}$ and $y = x^2$

xv). $x = \sqrt{y}$ and $y = 2x(1-x)$

xvi). $y = 2x(1-x)$ and $(1-x) = \sqrt{y}$

xvii). $f(x) = -x^2 + 4x$ and $g(x) = x^2$

xviii). $y = x(x-2) + 2$ and $f(x) = -x^2 + 6$.

xix). $y = x^2 - 1$ and $y + 1 = x^3$

xx). $y = x(x+3)$ and $f(x) = x(2x^2 - x - 5)$

323. On the same axes, sketch the given curve and the line, and in each case, calculate the exact area enclosed between the given curve and its line.

i). $x = \sqrt{y}$ and $y = x + 2$

ii). $f(x) = x^2$ and $g(x) = 2 - x$

iii). $y = 2x^2 + 10$ and $y = 4x + 16$

iv). $y = x(3-x)$ and $y = x$

v). $y = 3\sqrt{x}$ and $y = 3x$

vi). $y = x(x^2 - 9)$ and $x = \frac{1}{7}y$

vii). $\frac{y}{x} = 7 - 2x$ and $y = 3x$

viii). $x = \sqrt{y}$ and $y = x$

ix). $f(x) = x^2$ and $3x + 4 = g(x)$

x). $g(x) = x^3 + x^2 - x + 1$ and $f(x) = x + 1$

xi). $y = x^2 - 2x + 3$ and $x = 9 - y$

xii). $y = 4 - x^2$ and $x + y = 2$

xiii). $y = x(4 - x^2)$ and $y = 3$

xiv). $f(x) = 2 - x^2$ and $g(x) = x$

xv). $y = 6 - x$ and $y = 9 - \left(\frac{x}{2}\right)^2$

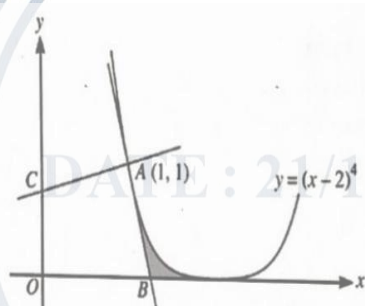
xvi). $y = 4x - x^2 - 1$ and $y = x + 1$

xvii). $y = x(5 - x)$ and $y - x = 0$

xviii). $y = -x^2 - 4x + 3$ and $y + 2x = 0$

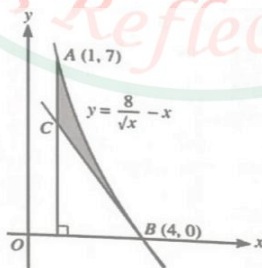
xix). $x = \sqrt{y}$ and $3y = 2x + 16$.

324. The diagram below shows the part of the curve; $y = (x - 2)^4$ and the point A(1,1) on the curve. The tangent at A cuts the x -axis at B and the normal at A cuts the x -axis at C.



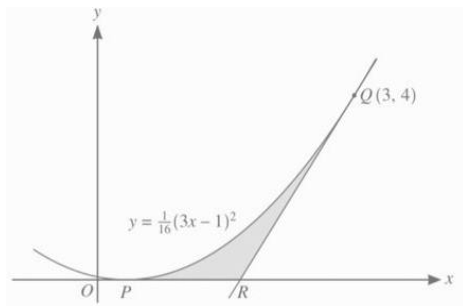
- Find the coordinates of B and C.
- Find the area of the shaded region.

325. The diagram below shows part of the curve; $y = \frac{8}{\sqrt{x}} - x$ at the points; A(1,7) and B(4,0) which lie on the curve. The tangent to the curve at B intersects the line $x = 1$ at point C.



- Find the coordinates of C
- Find the area of the shaded region.

326.



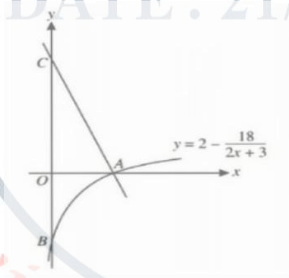
The diagram above shows the part of the curve; $y = \frac{1}{16}(3x - 1)^2$ which cuts the x -axis at point P. The point Q(3,4) lies on the curve and the tangent to the curve at Q crosses the x -axis at R.

- State the coordinates of point P
- Determine the exact area of the shaded region.

327. A curve; $y = 4 + 3x - x^2$ has a tangent; $l: x + y = 8$.

Show that the area of the region enclosed between; positive x -axis, the curve and the tangent, l is $13\frac{1}{3}$ square units.

328.

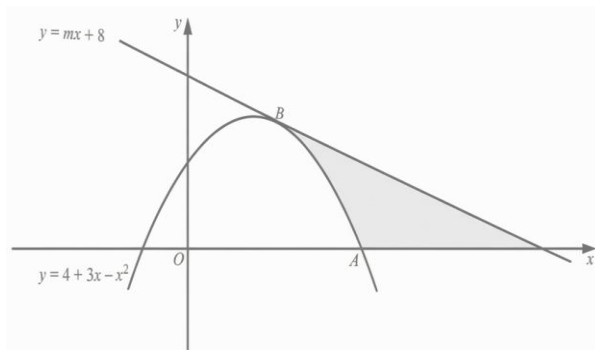


The figure above shows the part of the curve; $y = 2 - \frac{18}{2x+3}$.

The normal to the curve at A crosses the y -axis at C.

- Find the equation of line AC.
- Calculate the length of BC.

329.



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The diagram above shows the curve; $y = 4 + 3x - x^2$ intersecting the positive x -axis at point A. The line; $l: y = mx + 8$ is a tangent to the curve at point B.

Find;

- i). The coordinates of A
- ii). The value of m
- iii). The coordinates of B
- iv). The area of the shaded region.

CURVE SKETCHING II:

330. Sketch the curve; $y = \frac{3(x-3)}{(x+1)(x-2)}$.
331. A curve is parametrically given by; $x = 3t$ and $y = \frac{2t^2}{1-t}$.
 - i) Find the cartesian equation of the curve.
 - ii) Clearly show the region where the curve, y does not lie.
 - iii) Hence, find the stationary points.
 - iv) Sketch the curve
332. A curve, $f(x)$ is given by; $f(x) = \frac{(x-1)}{(2x-1)(x+1)}$.
 - i) Clearly, state the restricted region of $f(x)$.
 - ii) Hence, find the stationary points of $f(x)$.
 - iii) State the equations of asymptotes.
 - iv) Find the intercepts of the curve.
 - v) Hence, sketch the curve, $f(x)$.
333. Given the curve; $y = \frac{(3x-1)}{(4x-1)(x+5)}$.
 - i) Find the coordinates of any maxima, minima and points of inflexion y may have.
 - ii) Clearly, state the equations of the asymptotes of the curve.
 - iii) Hence, sketch the curve.
334. A curve is given by; $f(x) = \frac{3(x+1)}{x(3-x)}$
 - i) Find the intercepts of $f(x)$

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- ii) Clearly, state the equations of the asymptotes of $f(x)$.
 iii) Hence, sketch $f(x)$.
335. Given the curve; $f(x) = \frac{2(x-2)}{(x-1)(x-3)}$.
- i) Clearly, show that $f(x)$ has no stationary points.
 ii) Find the intercepts of $f(x)$
 iii) State the equations of asymptotes of $f(x)$
 iv) Hence, sketch the curve.
336. A curve, $f(x)$ is given by the parametric equations; $x = 4t$ and $\frac{3t^2}{(t-1)}$.
- i) Find the cartesian equation of the curve, $f(x)$.
 ii) Find the intercepts of $f(x)$
 iii) State the equations of the asymptotes of $f(x)$.
 iv) Hence, sketch $f(x)$
337. A curve is given by; $y = \frac{2(x-2)(x+2)}{(2x-5)}$.
- i) Determine the stationary points of the curve, and hence, the range of values of y for which the curve is undefined.
 ii) Determine the asymptotes of the curve.
 iii) Sketch the curve.
338. A curve is given by; $y = \frac{3x}{x^2-1}$.
- i) Find the stationary points of $f(x)$.
 ii) State the asymptotes of $f(x)$.
 iii) Sketch the curve, $f(x)$.
339. Given the curve; $y = \frac{x^2-3}{(x+2)}$.
- i) Find the range of values of y of the region from where the curve is restricted.
 ii) Find the asymptotes of the curve.
 iii) Hence, sketch the curve
340. Given the curve; $y = \frac{(x^2-4)}{(x^2-7x-18)}$
- i) Find the coordinates of the hole of the curve.
 ii) State the equations of the asymptotes of the curve
 iii) Find the intercepts of the curve
 iv) Sketch the curve.
341. A curve is given by; $h(x) = \frac{x^2+5x+6}{x^3-4x}$.
- i) State the coordinates of the hole of the curve.

- ii) State the equations of the asymptotes of $h(x)$.
 iii) Find the intercepts of $h(x)$.
 iv) Sketch the curve, $h(x)$.
342. A curve is given by; $f(x) = \frac{(3x-7)}{(x^2-2x-24)}$
 i) State the equations of the asymptotes of the curve
 ii) Find the turning points of $h(x)$
 iii) Sketch the curve.
343. A curve is given by; $y = \frac{(x^4-1)}{(x^3-x)}$
 i) State the coordinates of the hole.
 ii) Clearly, show that, for real values of x , the curve, y does not lie in the interval $|y| < 2$
 iii) Hence, find the stationary points of the curve, y .
 iv) State the equations of the asymptotes of the curve
 v) Hence, sketch the curve.
344. Given the curve; $y = \frac{4x}{x^3+8}$
 i) Find the coordinates of the intercepts.
 ii) Find the stationary points of the curve.
 iii) State the equations of the asymptotes of the curve.
 iv) Sketch the curve.
345. A curve is given by; $y = \frac{2x^2-9x-18}{x^2-x-2}$.
 i) Clearly, show the region where the curve does not lie
 ii) Hence, find the turning points of the curve
 iii) State the equations of the asymptotes
 iv) Find the intercepts of the curve
 v) Sketch the curve.
346. A curve is given by; $y = \frac{x^2}{(x-1)}$
 i) Clearly show the region where the curve lies.
 ii) Hence, find the stationary points of the curve
 iii) State the equations of the asymptotes of the curve.
 iv) Hence, sketch the curve.
347. Given the curve; $y = \frac{(x^2-4)}{(x-1)}$
 i) Clearly, find the equations of the asymptotes.

- ii) Find the intercepts of the curve.
iii) Hence, sketch the curve.
348. A curve is given by; $y = \frac{1+x^3}{4x}$
i) Determine the stationary points of the curve, if any.
ii) State the coordinates of the intercepts.
iii) Hence, sketch the curve.
349. A curve is given by; $f(x) = \frac{(x^2-3x+2)}{(x-3)}$. Another curve is given by; $h(x) = \frac{1}{f(x)}$.
Sketch the curves; $f(x)$ and $h(x)$ on the same axes.
350. Given the curve; $f(x) = \frac{x^3+3x^2+3x+1}{x^2-2x+1}$.
i) Determine the stationary points of the curve.
ii) Obtain the equations of the Asymptotes.
iii) Hence, sketch the curve.
351. A curve is given by; $y = \frac{4x+2}{x^2+4x-5}$.
i) Clearly, find the equations of the asymptotes of the curve.
ii) Find the coordinates of the points where the curve cuts x -axis and y -axis.
iii) Hence, sketch the curve.
352. Given that; $g(x) = \frac{x-1}{x^2-7x+12}$ and $f(x) = \frac{1}{g(x)}$.
Clearly, sketch both curves; $g(x)$ and $f(x)$ on the same axes.
353. A curve is given by; $y = \frac{2x^2-8}{2x-5}$
i). Show that for real values of x , y cannot take on the values in the interval;
 $2 < y < 8$.
ii). Find the nature of turning points
iii). Find the equations of the Asymptotes of the curve, hence, sketch the curve.
354. Given the curve; $y = \frac{3(x-1)}{(x+2)(3x-1)}$.
i). Determine the nature of all stationary points.
ii). Sketch the curve.
355. Given the curve; $y = \frac{2x^2-4}{2x-3}$
i). Show that for real values of x , y can not take values in the interval; $2 < y < 4$.
ii). Hence, obtain stationary points of the curve.
iii). State the equations of asymptotes.
iv). Hence, sketch the curve.

APPROXIMATIONS:

Under this, a learner may use the following standard Maclaurin's expansions.

(From Mathematical tables)

- i). $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots + \frac{x^n}{n!}$
- ii). $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \frac{x^9}{9!} + \dots + (-1)^n \frac{x^{2n+1}}{(2n+1)!} + \dots$
- iii). $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \frac{x^8}{8!} - \dots + (-1)^n \frac{x^{2n}}{(2n)!} + \dots$
- iv). $\log_e^{(1+x)} = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \dots + (-1)^{n+1} \frac{x^n}{n} + \dots$

356. Using Maclaurin's theorem, expand $e^{\frac{1}{3}x} \sin\left(\frac{1}{3}x\right)$ up to the term in x^3 .

Hence, evaluate $y = -3e^{\left(-\frac{\pi}{12}\right)} \sin\left(\frac{\pi}{12}\right)$ correct to three significant figures.

357. Given that; $y = \sin^{-1}(x)$.

Show that; $(1 - x^2) \frac{d^2y}{dx^2} = x \frac{dy}{dx}$. Hence, obtain the first two non – zero Maclaurin's terms of $f(x) = \frac{1}{4}y$

358. Obtain the first three non – zero Maclaurin's terms of; $(2 - 3x)^{\frac{1}{2}} \ln(3 + x^2)$.

Hence, find the approximate value of; $\ln 4$ correct to 4 decimal places.

359. Find the Maclaurin's series for; $(1 + x)^{-1}$ as far as x^3 .

Hence, deduce the Maclaurin's series for;

- i). $\frac{1}{1+x^2}$ as far as x^6
- ii). $\tan^{-1}(x)$ as far as x^7

360. Use Maclaurin's theorem to expand; $\ln\left\{\sqrt{\frac{(1+2x)}{(1-3x)}}\right\}$ up to the term in x^4 .

Hence, evaluate;

- i). $\ln\{\sqrt[3]{4}\}$,
- ii). $\ln\{\sqrt[5]{3}\}$,
- iii). $\ln\left(\frac{1}{4}\right)$, correct to 4 decimal places.

361. Obtain the first three non – zero Maclaurin's terms of; $\cos\left(\frac{3}{4}x\right)$.

Hence, deduce the first three non – zero Maclaurin's terms of;

- i). $\sin^2\left(\frac{3}{8}x\right)$

ii). $\cos^2\left(\frac{3}{8}x\right)$

362. Find the Maclaurin's series expansion for function; $f(x) = e^{-x^2}$ up to the term in x^4 . Hence, deduce that; $e^{2x-x^2} \cong 1 + 2x + x^2 - \frac{2}{3}x^3$.
363. Given that; $y = e^{\tan^{-1}(x)}$. Show that; $(1+x^2)\frac{d^2y}{dx^2} = (1-2x)\frac{dy}{dx}$
Hence or otherwise, determine the first four non – zero terms of the Maclaurin's expansion of y .
364. Obtain the first three non – zero Maclaurin's terms of; $\ln(1 + \sin x)$
365. Given that; $f(x) = \frac{1}{\sqrt{1+x}}$.
Obtain the first four non – zero Maclaurin's terms of $f(x)$, and show that;
 $f(x) \cong 1 - \frac{x}{2} + \frac{3}{8}x^2 - \frac{5}{16}x^3$. Hence, find the approximate value of $\frac{2}{\sqrt{1.04}}$ correct to 4 decimal places.
366. Given that; $f(x) = e^{x^2}(\sqrt{1+2x})$.
By expanding, Using Maclaurin's theorem, up to the term in x^3 , show that;
 $f(x) = 1 + x + \frac{1}{2}x^2 + \frac{3}{2}x^3 + \dots$
367. Expand $e^{\sin x}$ by Maclaurin's theorem as far as the term involving x^4 .
368. Given that; $f(x) = \sqrt[3]{e^{2x}\cos x}$.
By expanding using Maclaurin's theorem, show that; $f(x) \cong 1 - \frac{5}{3}x + \frac{31}{18}x^2$.
369. Find the Maclaurin's expansion of; $\ln\left[\sqrt[3]{\frac{1+2x}{1-2x}}\right]$ up to and including the term in x^3 .
370. Given that; $g(x) = \frac{1+e^x}{2e^{2x}}$.
By Maclaurin's expansion, show that; $g(x) \approx 1 + \frac{1}{8}x^2 + \frac{1}{384}x^4$.

END:

“IF YOU CAN'T EXPLAIN IT IN THE VERY SIMPLEST FORM, THEN YOU DON'T KNOW IT”

PROVERBS 13:04